

Week 5 Kigali Weather Report

AUCA · Introduction to Big Data Analytics

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Dataset: 60 days of Kigali weather, September–October 2026.

Key findings

- Mean temperature: 24.00 °C; maximum: 28.6 °C on 2026-09-17; minimum: 19.3 °C.
- Hot days (≥ 26 °C): 18. Rainy days: 16. Total rainfall: 331.2 mm. Heaviest rainfall: 26.2 mm.
- Average temperature on rainy days: 24.14 °C. On dry days: 23.95 °C.
- First 30-day average: 26.21 °C. Last 30-day average: 21.78 °C. Change: -4.43 °C.

Insight

The clearest pattern is a cooling trend. The last 30 days were about 4.43 °C cooler on average than the first 30 days. By comparison, the rainy-versus-dry temperature difference was only about 0.19 °C. Therefore, the time trend is more prominent in this dataset than rainfall status alone.

Conclusion

NumPy simplified the analysis through array methods, vectorized calculations, and boolean masks. These tools made it possible to summarize the 60-day Kigali dataset without writing explicit loops in the required NumPy sections.